S2.2 The covalent model

Intermolecular forces

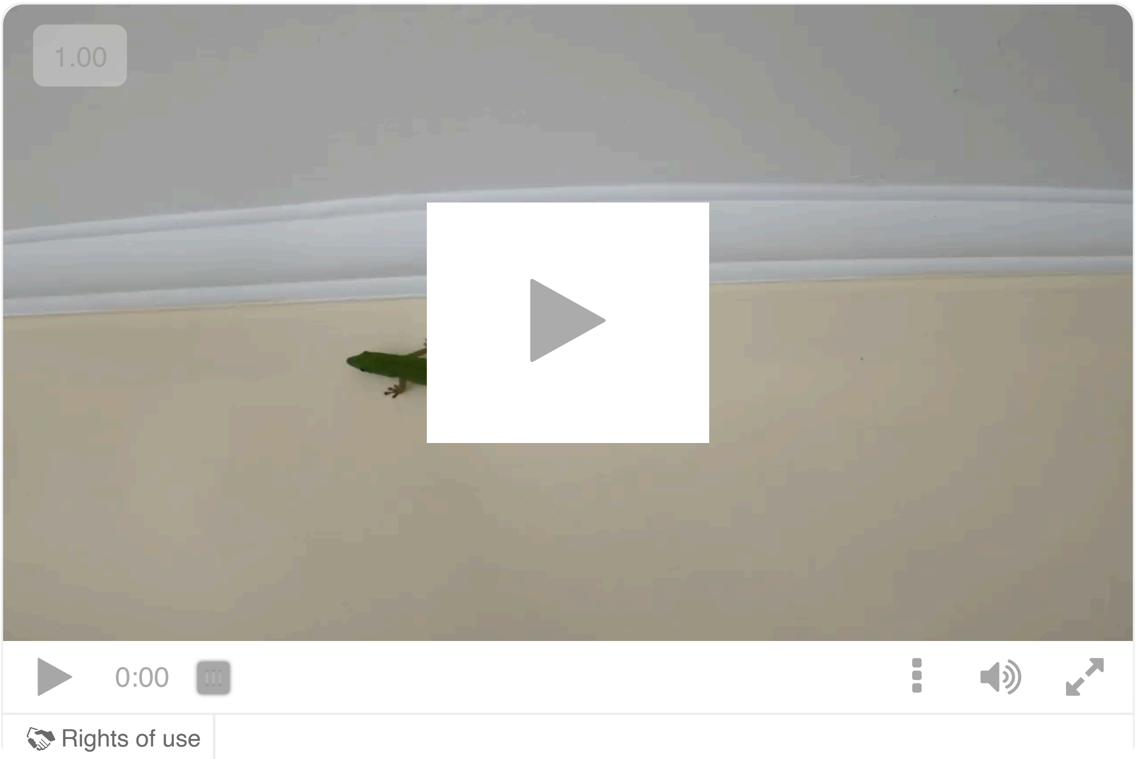
Learning outcomes

By the end of this section you should be able to:

- Identify the types of intermolecular forces present in molecules as dipole—dipole, dipole-induced-dipole and London (dispersion) forces, encompassed by the umbrella term 'van der Waals forces'.
- Deduce the type(s) of intermolecular force present from the electronegativity difference and structural features of molecular compounds.
- Describe the main features of each type of intermolecular force.

Geckos, reptiles found on every continent in the world except Antarctica, have an amazing ability to run up smooth walls and hang upside down off ceilings without falling off (**Video 1**). They can do this without having any kind of sticky substance or suction cups on their feet. How is this possible?

Scientists have recently discovered the phenomenon that allows them to do this. It is based on their ability to maximise the attraction their feet can form with the particles on the surface they are climbing up.



Video 1. Geckos Found on Every Continent Except Antarctica.

More information for video 1

An interactive video shows a green-colored gecko moving on a vertical wall near the ceiling. The gecko has a slender body and a long tail. It has four limbs, each with specialized toe pads that enable it to climb vertical surfaces. The gecko moves using its four limbs, where each forelimb moves in sync with the opposite hindlimb.
This video helps users understand the structure of geckos and their ability to move on walls.

Intermolecular forces versus intramolecular forces

So far in this subtopic, we have looked at how the atoms within individual molecules interact with each other to create bonds. As these bonds occur within the molecule, these forces are known as an intramolecular, meaning within the molecule. Ionic and covalent bonds are examples of intramolecular forces. Metallic bonds are another type of intramolecular force that you will learn about in subtopic S2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43111/).

Molecules are very, very small and even a small item you use in daily life is probably made up of trillions of molecules. For example, a small glass of water (approximately 200 cm^3) is made up of 7 septillion water molecules: that's

7000000000000000000000000 (7×10^{24}) molecules!

Have you ever wondered what happens when so many molecules make up a sample? How do these molecules interact with each other? As these forces would be occurring between molecules rather than within them, a new term is needed to describe them: intermolecular forces (**Figure 1**).

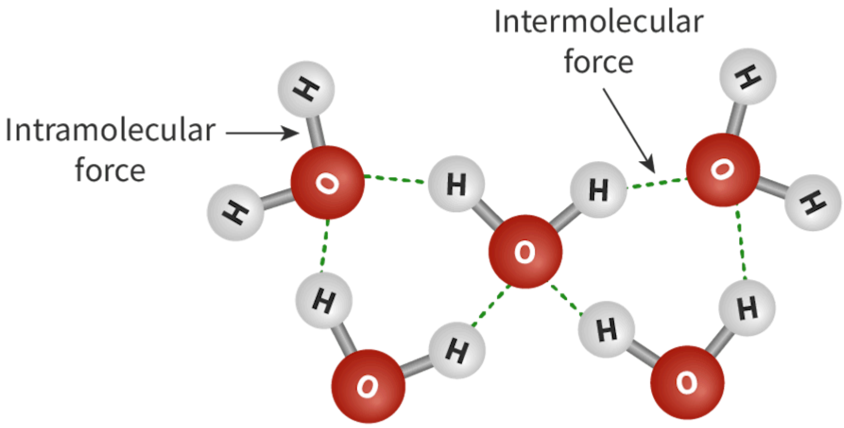


Figure 1. Intermolecular and intramolecular forces.

More information for figure 1

The image is a diagram illustrating the concepts of intermolecular and intramolecular forces. It depicts two molecules with different types of forces. Each molecule consists of a central oxygen atom (labeled 'O') connected to two hydrogen atoms (labeled 'H'), forming a V-shape. This arrangement represents water molecules.

Intramolecular forces are indicated by solid lines connecting the oxygen atom to the hydrogen atoms within the same molecule. These represent the covalent bonds holding the atoms together within a molecule.

Intermolecular forces are depicted as dashed lines between the hydrogen atoms of one molecule and the oxygen atom of an adjacent molecule. These lines illustrate the hydrogen bonds that occur between molecules, which are responsible for holding different molecules together.

Labels are present on the diagram to indicate 'Intramolecular force' with an arrow pointing to the solid line within a molecule and 'Intermolecular force' with an arrow pointing to the dashed line between molecules.

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The bromine molecule demonstrates both intermolecular and intramolecular forces in action.

Bromine, Br_2, is a liquid at room temperature. Covalent bonds hold the bromine atoms together within the molecule: the intramolecular forces (**Figure 2**). However, there must be some other forces of attraction acting between them that keep the bromine molecules as liquid molecules, otherwise these molecules would separate completely and act as gases.

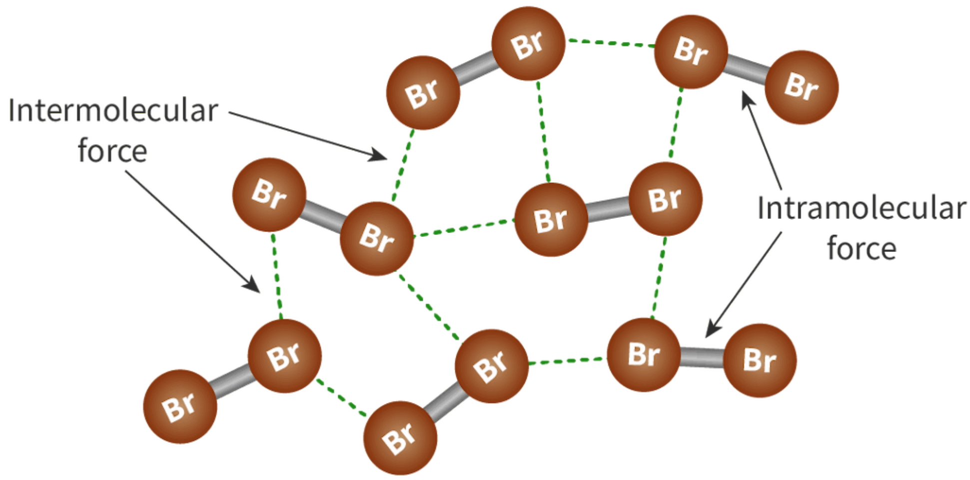


Figure 2. Intermolecular and intramolecular forces of bromine molecules.

More information for figure 2

The diagram illustrates bromine molecules (Br2) highlighting the forces acting within and between them. Each bromine molecule is represented as two brown "Br" circles connected by a solid gray line, indicating covalent bonds or intramolecular forces. These bonds hold individual bromine atoms together within a molecule. Dotted green lines connect different bromine molecules, representing intermolecular forces acting between them. Two black arrows label and point to these forces: "Intramolecular force" points to the solid gray lines, and "Intermolecular force" points to the dotted green lines. This visual demonstrates how these forces keep bromine in liquid form at room temperature by holding the discrete molecules together.

[Generated by AI]

The forces acting between discrete molecules of bromine are intermolecular forces and they are responsible for the physical state of matter of covalent substances, i.e. if they are solid, liquid or gas. When you melt or boil a substance it is the intermolecular forces that must be overcome, not the intramolecular forces. Intramolecular forces are much stronger than intermolecular forces. How could this be used to explain why the melting point of a substance with a covalent network structure is so much higher than that of a simple molecular compound?

Study skills

Many people mistakenly think that when a substance changes state (e.g. changes from liquid to gas) that the bonds within the molecule are breaking. This is not the case! The bonds within the molecule stay intact, it is the forces between molecules that are being overcome.

Think about when you boil some water; the steam produced (gaseous water) is still water: it has not split into hydrogen and oxygen atoms (**Figure 3**).

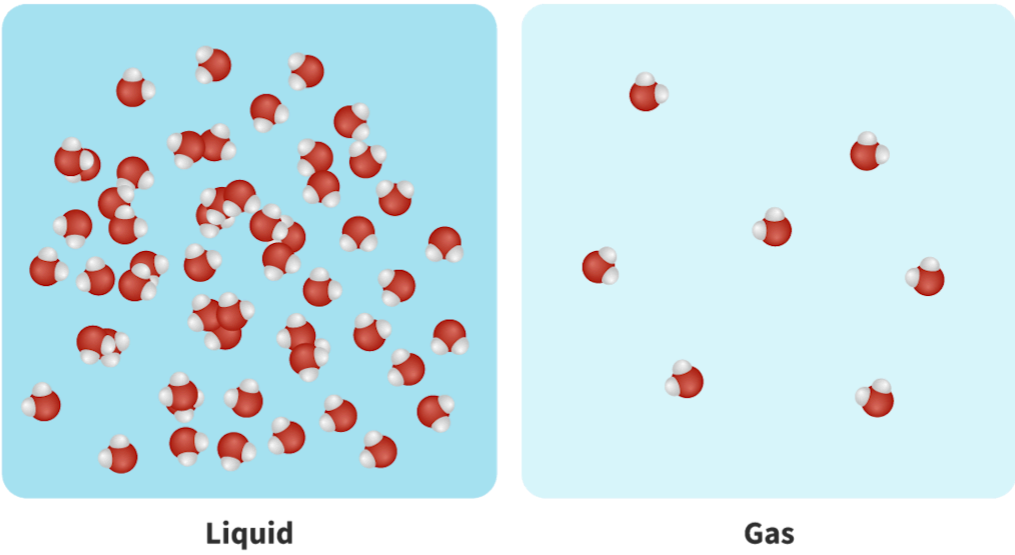


Figure 3. Water as liquid and as gas.

Remember that there can be electrostatic forces of attraction within the molecule with regions of partial positive and negative charges, which can form dipoles when there is a difference in their electronegativities. How might that affect the way the particles interact and arrange themselves in a sample?

Let’s examine two other molecules, ethoxyethane and butanol. These molecules have identical molecular masses but very different boiling points (**Figure 4**). What determines boiling points and why are they are so different?

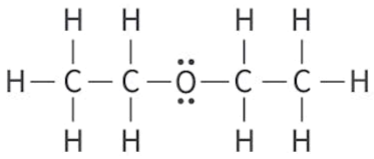
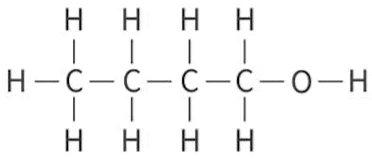
	Ethoxyethane	Butanol
Lewis formula :		
Molecular mass (g mol^{-1}) :	74.14	74.14
Boiling point (°C) :	35°C	117°C

Figure 4. Identical molecular masses of two molecules with different boiling points.

More information for figure 4

The image is a comparison table of two molecules, ethoxyethane and butanol. It consists of three main rows: Lewis formula, molecular mass, and boiling point, each labeled on the left.

Underneath the label 'Ethoxyethane,' the Lewis formula is shown with two carbon atoms bonded to an oxygen atom (C-O-C), with hydrogen atoms filling the remaining bonds. The molecular mass for ethoxyethane is listed as 74.14 g/mol. The boiling point is 35°C.

Under 'Butanol,' the Lewis formula shows a straight chain of carbon atoms (C-C-C-C) with the end carbon bonded to a hydroxyl group (O-H), and hydrogen atoms completing the remaining bonds. Butanol also has a molecular mass of 74.14 g/mol, but its boiling point is significantly higher at 117°C.

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Recall from section S1.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) that the physical state of matter (solid, liquid or gas) is determined by the melting and boiling points. The difference in boiling point for these two molecules, ethoxyethane and butanol, indicates all intermolecular forces must not be the same. What are the different types of intermolecular force? What factors determine the type of intermolecular force that will form?

Types of intermolecular force

The types of intermolecular force are:

- London dispersion
- Dipole-induced dipole
- Dipole–dipole
- Hydrogen bonding

Concept

Of the four types of intermolecular forces, three of them can be considered under an umbrella term of van der Waals forces:

- London dispersion
- Dipole-induced dipole
- Dipole—dipole

Hydrogen bonding is not considered to be one of the van der Waals forces.

However, when answering questions, you should also refer to the specific type of force rather than using the umbrella term.

In **Interactive 1,** drag and drop the information into the table to see the differences between the two types of forces.

	Intramolecular	Intermolecular
Type of force		
Diagram		
Examples		
Strength		

Covalent

Stronger

Between molecules

Dipole-dipole

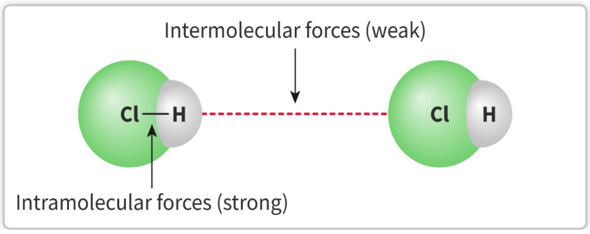
Weaker

Within molecules

Hydrogen bonding

Ionic

London dispersion



Check

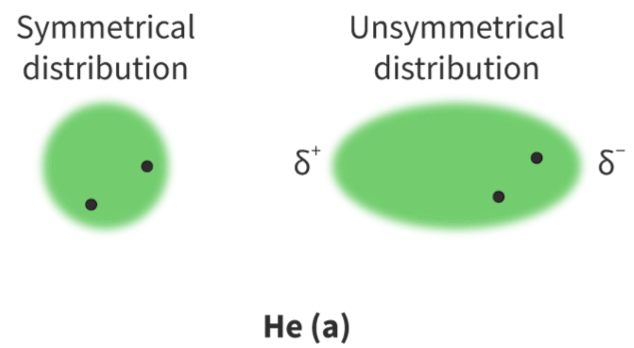
Interactive 1. Differences Between Intra- and Intermolecular Forces.

London dispersion forces

Consider a bond between two atoms that has a very low or non-existent electronegativity difference. How would you expect the electrons to be distributed in this non-polar bond?

You may expect that the electrons would be found roughly in the middle of these two bonded atoms when atoms have similar electronegativity values. However, recall that electrons are not stationary: they are constantly moving, in a completely random direction, even in nonpolar bonds. They are sometimes found temporarily in the middle of a bond between two atoms... but this is only temporary. A split second later, those same electrons may be found slightly closer to one side than the other. And seconds later, this could be reversed.

Therefore, while we describe these bonds as nonpolar, indicating that these bonds have symmetrical electron distribution: this is only on **average.**



More information

The image is a diagram demonstrating the concept of electron distribution in helium. It shows two helium atoms, one labeled 'Symmetrical distribution' and the other 'Unsymmetrical distribution'.

- The symmetrical distribution is depicted with a smaller, circular green shape symbolizing electrons that are evenly distributed, with two small black dots inside the shape representing electrons.
- The unsymmetrical distribution is illustrated with an elongated green shape, indicating an uneven spread of electrons. There are also two black dots, but they are positioned to show a relative imbalance. Next to this shape, the notation δ+ is on one side and δ− on the opposite side, highlighting areas of partial positive and negative charge. The text 'He (a)' is centered below the diagram.

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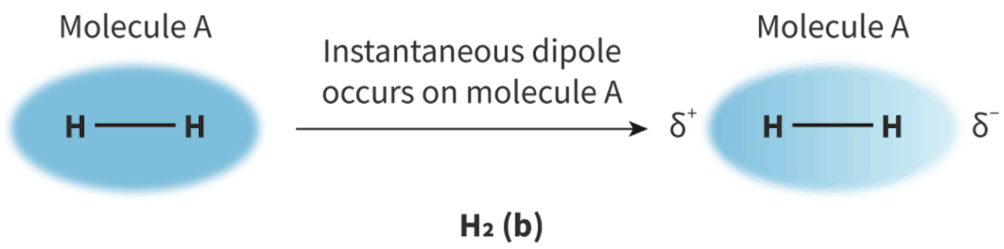

Figure 5. London dispersion forces: the random, continuous movement of electrons can create asymmetrical electron distribution which causes attraction between neighbouring atoms or molecules. This can occur in an atom as in helium (a) or a molecule as in hydrogen (b).

More information for figure 5

The image is an illustration of a hydrogen molecule (H_2) depicted twice as 'Molecule A'. On the left side, a blue oval labeled 'Molecule A' contains two hydrogen atoms (H) connected by a bond line. A text in the middle explains that an "instantaneous dipole occurs on molecule A", with an arrow pointing to the right. This suggests a change in electron distribution. On the right side, a similar blue oval represents the same 'Molecule A' after the dipole formation, showing partial positive charge (δ^+) on one hydrogen and partial negative charge (δ^-) on the other, indicating an asymmetrical electron distribution.

[Generated by AI]

You will recognise the same symbols in **Figure 5** of the molecules that you used to represent charges on a polar bond. Note that these are only representative of electron distribution at a singular moment in time, rather than on a continuous basis as with polar bonds.

An instant later, the electrons may redistribute and reverse the ‘polarity’ of the molecule. This causes rapidly fluctuating dipoles even in molecules that are completely symmetrical. Can you see why these are sometimes referred to as temporary or instantaneous dipoles?

Making connections

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To what extent can intermolecular forces explain the deviation of real gases from ideal behaviour (subtopic S1.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-44341/)).

Look back to the bromine molecule, Br_2. The symmetrical bromine molecule will appear, on average, to have no net dipole. However, the mobility of these electrons means that they can randomly end up at one side of the molecule and temporarily make it δ−. As the other end of the molecule is then short of electrons it becomes δ+. Use **Figure 6** to help you visualise this.

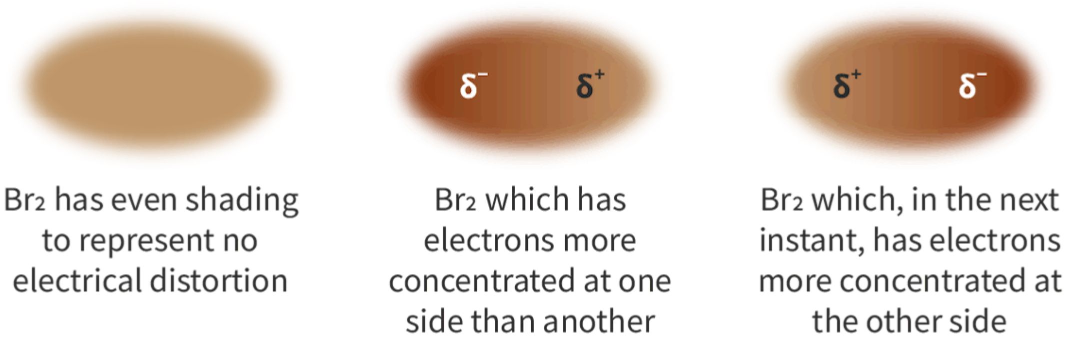

Figure 6. London dispersion forces in a symmetrical bromine molecule.

More information for figure 6

The diagram illustrates the process of electron mobility in a bromine (Br) molecule through three stages. On the left, the Br molecule is depicted with even shading, indicating no electrical distortion, accompanied by the text: "Br has even shading to represent no electrical distortion." In the middle, the molecule shows concentrated electrons on one side, marked with **δ**− and **δ**+ symbols to indicate a temporary dipole, with the text reading: "Br which has electrons more concentrated at one side than another." On the right, the molecule illustrates a shift where the electrons are concentrated on the opposite side, again marked with **δ**+ and **δ**− symbols, and the text states: "Br which, in the next instant, has electrons more concentrated at the other side." This visualizes the transient nature of electron distribution and the formation of dipoles.

[Generated by AI]

The presence of these temporary dipoles is what causes attraction between neighbouring molecules that both experience this asymmetry in electron distribution.

London dispersion forces exist between **all** molecules, regardless of whether the electronegativity difference is very low or non-existent and even if other forces are also acting. They are the weakest type of intermolecular force.

At the beginning of this section, we identified that geckos have a remarkable ability to run across walls and hang upside down from ceilings. They achieve this through the London dispersion forces that exist between their feet and the walls (**Figure 7**).

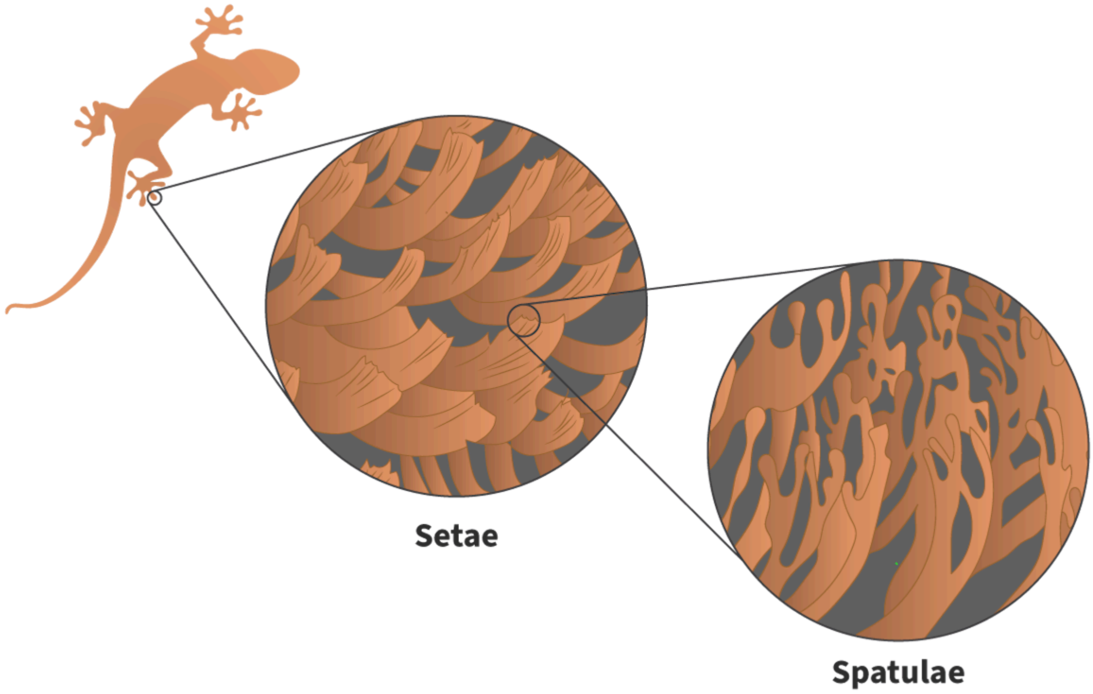

Figure 7. The structure of a gecko's feet maximises the London dispersion forces between them and the surfaces they climb. Geckos can maximise the impact of these forces by increasing the surface area of their feet. Their toes have a huge number of tiny hairs (setae) that further branch into lots of small triangular tips (spatulae). Their combined effect of the London dispersion forces over millions of these tiny spatulae is enough to support the weight of the gecko.

More information for figure 7

The image is a diagram illustrating the structure of a gecko's foot, highlighting the intricate details that allow it to adhere to surfaces. On the left, there is a silhouette of a gecko. This is connected by a line to a magnified view of setae, which appear as numerous hair-like structures. Further magnified, these setae are shown branching into smaller, triangular tips known as spatulae. The spatulae are extensively detailed, emphasizing their role in maximizing surface contact. Labels indicate 'Setae' and 'Spatulae', clarifying the hierarchical structure in the gecko's foot anatomy.

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[Generated by AI]

Dipole—dipole

Recall from section S2.2.6a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molecule-polarity-id-45130/) that if polar bonds are not arranged symmetrically in a molecule then the molecule will have a permanent dipole. This explains why a different intermolecular force exists in asymmetrical molecules.

The hydrogen bromide molecule, HBr, in **Figure 8** is polar. This means it has regions of the molecule that have permanent partial charges. What do you think would happen if two or more hydrogen bromide molecules were near each other?

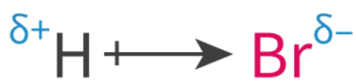

Figure 8. In a hydrogen-bromine bond, the bromine pulls electrons closer to it and gains a **δ**− negative charge, and the hydrogen gains a **δ**+ charge.

More information for figure 8

This diagram illustrates a hydrogen-bromine (HBr) bond, representing the polar nature of the molecule. On the left side, the letter 'H' indicates a hydrogen atom with a partial positive charge, denoted as 'δ+'. An arrow points from the hydrogen towards the right, directing attention to the 'Br', representing a bromine atom. The bromine is labeled with a 'δ-' to indicate its partial negative charge. The arrow suggests the movement of electrons towards the bromine atom, explaining the charge distribution. Overall, the image demonstrates the concept of polarity where bromine attracts electrons more strongly than hydrogen, leading to a partial negative charge on bromine and a partial positive charge on hydrogen.

[Generated by AI]

When two hydrogen bromide molecules are near each other, the opposite partial charges on the molecules are attracted to each other. You can think of them as acting like magnets, the north pole of one magnet will attract the south pole of a neighbouring magnet and vice versa.

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This generates a different type of intermolecular force known as a dipole–dipole attraction.

This type of force will only exist between two asymmetrical molecules that each have a permanent dipole. Dipole–dipole forces are much stronger than London dispersion forces, but can act in unison with them (**Figure 9**).

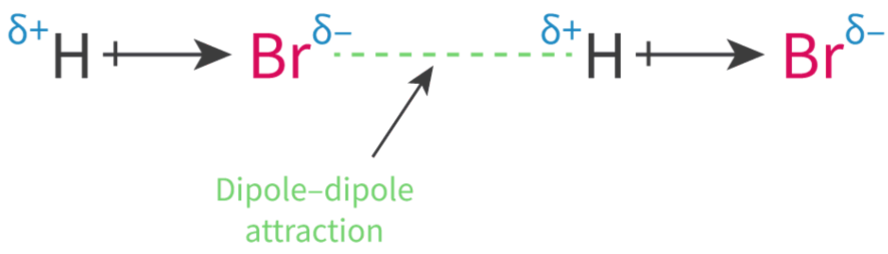

Figure 9. In hydrogen bromide molecules, the **δ**− on one HBr molecule attracts the **δ**+ on a neighbouring HBr molecule.

More information for figure 9

The diagram illustrates the dipole-dipole attraction between hydrogen bromide (HBr) molecules. It shows two HBr molecules aligned linearly. Each hydrogen atom (H) is labeled with a δ+ (delta plus), indicating a partial positive charge, while each bromine atom (Br) is labeled δ− (delta minus), indicating a partial negative charge. An arrow points from the δ+ of one H atom towards the δ− of the Br atom in the neighboring molecule, illustrating the attraction. The text 'Dipole-dipole attraction' is annotated alongside the arrow. The diagram visually communicates the interaction between the differentially charged ends of the molecules, which results in the dipole-dipole attraction.

[Generated by AI]

Dipole-induced-dipole

Dipole–dipole forces exist in asymmetrical molecules that have a permanent dipole. The partial positive on one molecule attracts the partial negative on another molecule and vice versa. What do you think would happen if a very polar molecule approached a molecule which is non-polar?

When a polar molecule approaches an atom or a non-polar molecule, the partial charges on the polar molecule can affect the distribution of electron density in the neighbouring atom or molecule (**Figure 10**). This causes the

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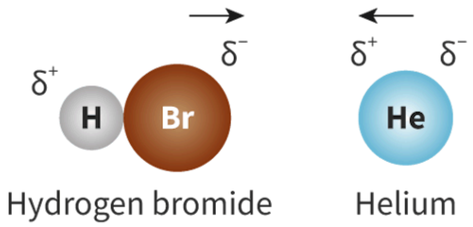
formation of a temporary dipole on the non-polar species. It is temporary as it is completely dependent on it being next to the polar molecule.

This results in a new type of intermolecular force known as a dipole-induced-dipole force.



More information

The diagram illustrates a dipole-induced dipole force between hydrogen bromide and helium. On the left, there is a molecule of hydrogen bromide, depicted as two circles: one labeled 'H' with a δ+ symbol and a smaller particle, and one labeled 'Br' with a δ- symbol and a larger particle. An arrow pointing towards the right indicates the direction of the dipole.

To the right, there is a circle labeled 'He' representing helium, with a δ+ and δ- symbol indicating an induced dipole. An arrow pointing towards the left shows the direction of the induced polarization in helium.

The text below the circles reads 'Hydrogen bromide' for the left side and 'Helium' for the right side, showing the interaction between them.

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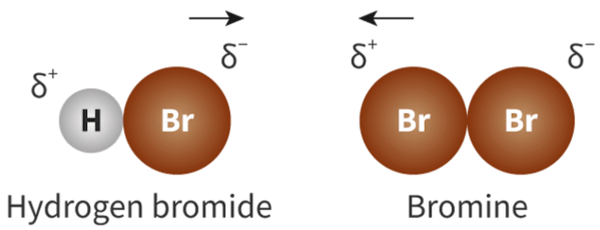

Figure 10. Polar hydrogen bromide molecule can induce a dipole in an atom (e.g. He) or non-polar molecule (e.g. Br_2).

More information for figure 10

The image is a diagram illustrating the interaction between a polar hydrogen bromide (HBr) molecule and a bromine (Br2) molecule. On the left side, the hydrogen bromide molecule consists of two circles: a smaller grey circle representing hydrogen (H) with a δ+ label, and a larger orange circle representing bromine (Br) with a δ− label. An arrow points from the hydrogen toward the bromine to indicate the direction of the dipole moment. To the right, there are two slightly overlapping larger orange circles, each labeled "Br," representing a bromine molecule. Between these two circles is a label showing δ+ on the side closer to the hydrogen bromide and δ− on the side farther from it, indicating the induced dipole within the bromine molecule. Below the illustration, text labels identify the molecules as "Hydrogen bromide" and "Bromine." The arrows between the circles indicate the directional polarity and interaction between the molecules.

[Generated by AI]

How strong this dipole-induced-dipole force is will depend on how susceptible the non-polar molecule is to an electric field, i.e. the δ– and δ+ of neighbouring molecule. Large molecules are more susceptible than smaller ones as they have larger electron clouds and the electrons are not being held as tightly by the nucleus. This results in greater scope for distortion.

In this next activity, you will identify the intermolecular forces acting for different molecules. Remember that to determine whether a molecule will experience dipole–dipole forces, the overall molecule must be polar and that depends on the presence of polar bonds and the symmetry on the molecule.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 25 minutes
- **Activity type:** Individual activity

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Draw the Lewis formula for each of the molecules below and identify **all** intermolecular forces present.

1. PBr_3
2. ICl
3. I_2
4. CH_2F_2
5. C_4H_{10}
6. Ne
7. PCl_3
8. F_2
9. CO_2
10. H_2S

S2.2 The covalent model

Hydrogen bonding

Learning outcomes

By the end of this section you should be able to describe hydrogen bonding as a type of intermolecular force present in some molecules based on the structural features of molecular compounds.

The boiling point of a series of group 16 elements bonded to hydrogen, known as the group 16 hydrides, was measured. The graph in **Figure 1** was drawn to represent this data.

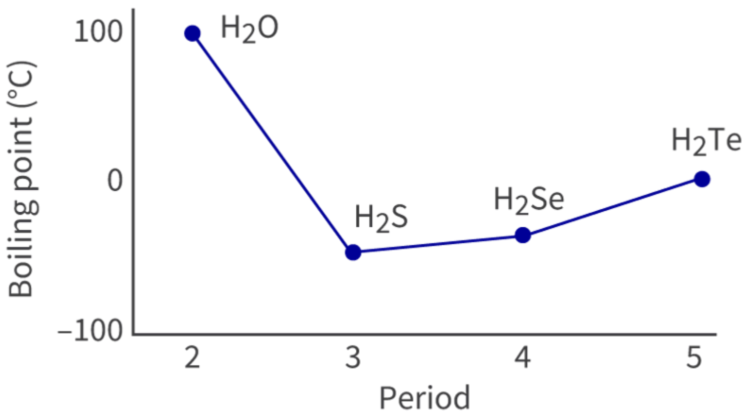


Figure 1. Boiling point of group 16 hydrides.

More information for figure 1

The graph illustrates the boiling points of group 16 hydrides, specifically H2O, H2S, H2Se, and H2Te. The X-axis represents the Period, marked from 2 to 5, while the Y-axis shows the Boiling Point in degrees Celsius, ranging from -100 to 100. \n\nData points are connected by a line and marked for each hydride: H2O has a boiling point of 100°C in Period 2, dramatically dropping to around -60°C for H2S in Period 3. This low value remains similar for H2Se in Period 4, and then increases slightly to around -40°C for H2Te in Period 5. The trend highlights a spike in boiling point for H2O compared to others, indicating a deviation from the general trend.

[Generated by AI]

The elements in periods 3, 4 and 5 follow a clear trend but the period 2 hydride, water, does not. What feature of the hydrogen molecule is responsible for causing such an increase?

Hydrogen bonding

Water is an example of a molecule that experiences a different type of intermolecular force known as hydrogen bonding (**Figure 2**).

This intermolecular force only exists in molecules that have a bond between a hydrogen and a very electronegative element such as fluorine, oxygen or nitrogen (sometimes nicknamed 'FON' for short), and the electronegative element must have a non-bonding pair of electrons.

Requirements for a hydrogen bond:

- Hydrogen bonded to a FON element.
- The FON element must have a non-bonding pair of electrons.

The hydrogen bond is the attraction for the δ+ hydrogen and the non-bonding pair of electrons on a FON element in a nearby molecule (atom B in **Figure 2**). The hydrogen bond is **not** the intramolecular bond that exists between hydrogen and the FON in the same molecule.

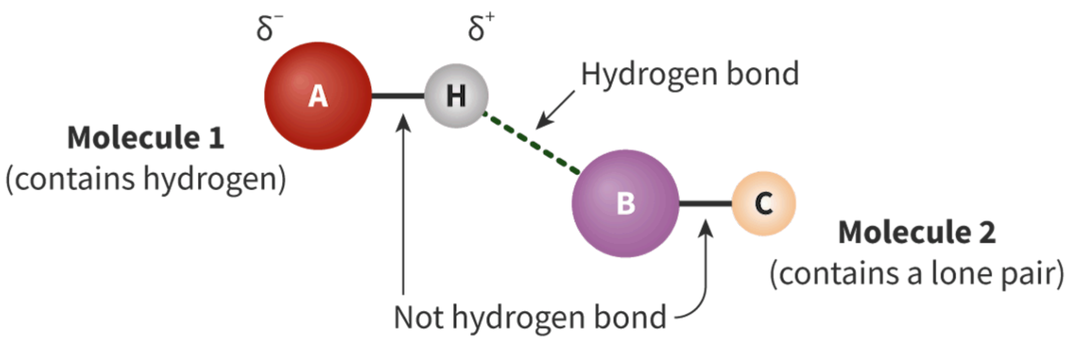


Figure 2. Hydrogen bonding.

More information for figure 2

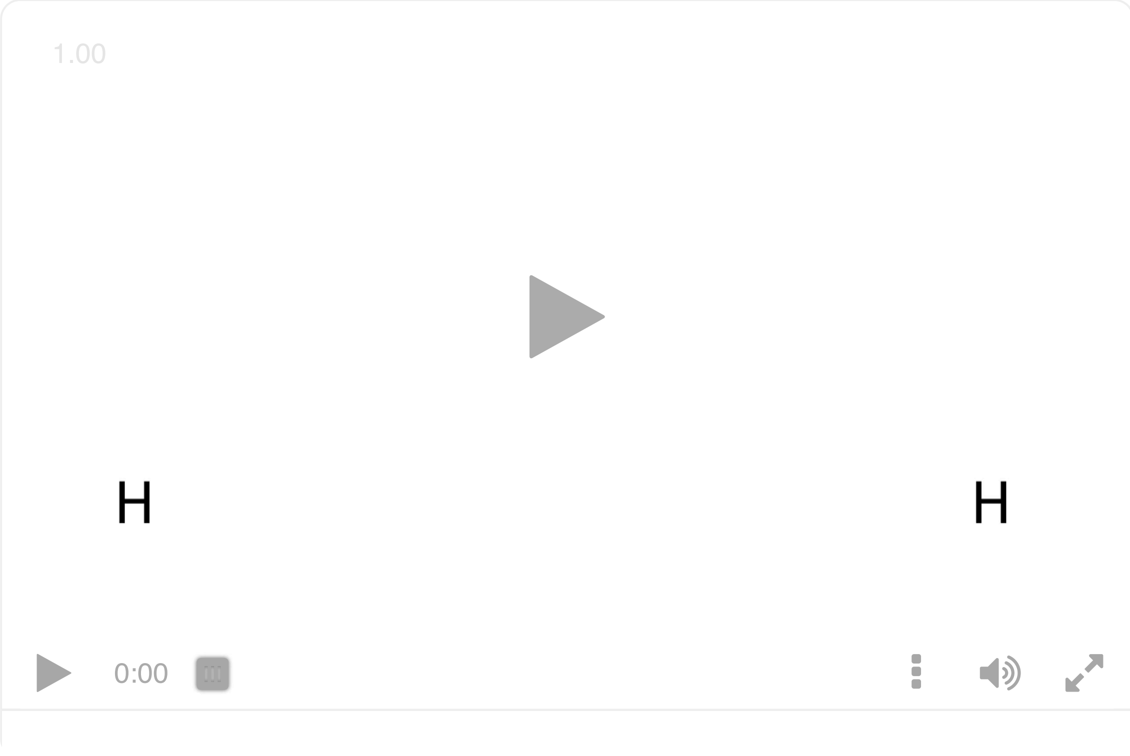
The diagram depicts the interaction of hydrogen bonds between two molecules, labeled as Molecule 1 and Molecule 2. Molecule 1 consists of three atoms: A, H, and another labeled δ-. Atom A is depicted as a large red sphere with the label 'A' and is part of Molecule 1, which 'contains hydrogen.' The H atom is shown as a smaller, gray sphere labeled 'H,' positioned between A and another atom B, which is part of the adjacent Molecule 2. The B atom is a large purple sphere labeled 'B,' representing an atom with a lone pair. A green dotted line labeled 'Hydrogen bond' connects H and B, illustrating the formation of a hydrogen bond between a δ+ hydrogen atom and a donor atom with a lone pair of electrons. On the left side, text indicates that the bond between atoms A and H is “Not hydrogen bond.” The diagram shows the partial charges, with A labeled as δ- (negative) and H as δ+ (positive).

[Generated by AI]

What is so special about the intramolecular bond between hydrogen and the FON element that it can result in such a strong dipole?

There is a very significant electronegativity difference between these elements. This allows the hydrogen to have a strong δ+. The strength of this δ+ is increased by the fact that hydrogen has no inner electrons to shield the impact of its nucleus. This makes the δ+ hydrogen very attracted to non-bonding pairs of electrons on neighbouring molecules.

Consider a molecule such as water in **Interactive 1**. Water molecules can form hydrogen bonds with other neighbouring water molecules.



Interactive 1. Hydrogen Bonding in Water Molecules.

More information for interactive 1

A video depicts the hydrogen bonding between molecules of water. Two H atoms move towards the same O atom and bond with it by a single bond. O has two pairs of electrons. The bond between the O and H atoms is labeled as "intramolecular polar covalent bond". O has a delta negative charge, and both atoms have a delta positive charge. Another H_2O molecule is placed on the top right of the central molecule. The O atom of the first molecule and one of the H atoms of the second molecule are bonded and labeled by "hydrogen bond". The third H_2O molecule is placed on the top left and the hydrogen bond is created between the O of the first molecule and the H atom of the third molecule.

Hydrogen bonding is the strongest intermolecular force. Recall the graph at the beginning of this section. It showed that the boiling point of water was so much higher than the other group 16 hydrides. Why can the other group 16 hydrides not form hydrogen bonds in the same way that water can?

Nature of Science

Aspect: Evidence

Over the last century, the definition of what is meant by a hydrogen bond has changed as more experimental and theoretical evidence becomes available.

Some scientists predict that the current accepted IUPAC (International Union of Pure and Applied Chemistry) definition will undergo another revision in the next 30 years as technology advances and more evidence can be discovered.

Hydrogen bonding can happen between two molecules that are the same, as with water molecules above, or two molecules that are different (**Figure 3**). However, both molecules must satisfy the requirements of the hydrogen bond: having hydrogen bonded to a highly electronegative element such as fluorine, oxygen or nitrogen **and** the electronegative element also having a non-bonding pair of electrons.

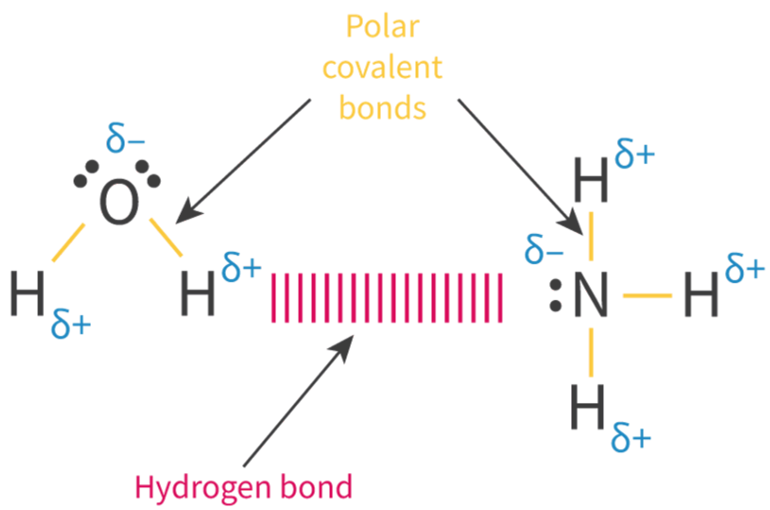


Figure 3. Hydrogen bonding can happen in two molecules that are different, for example water and ammonia.

More information for figure 3

The diagram shows the hydrogen bonding between a water molecule (H_2O) and an ammonia molecule (NH_3). On the left side, the water molecule is depicted with a central oxygen atom (O) bonded to two hydrogen atoms (H) with polar covalent bonds. The diagram uses blue delta symbols (δ-) to indicate the partial negative charge on the oxygen and blue delta symbols (δ+) on the hydrogens to indicate partial positive charges.

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On the right, the ammonia molecule is shown with a central nitrogen atom (N) bonded to three hydrogen atoms. The nitrogen has a partial negative charge (δ-) and each hydrogen has a partial positive charge (δ+), marked similarly with the blue delta symbols.

Between the hydrogen of the water molecule and the nitrogen of the ammonia, a hydrogen bond is indicated by a series of dashed lines in magenta. Arrows labeled "Polar covalent bonds" point from the nitrogen and oxygen towards their respective hydrogens, showing the direction of the bond polarity. An arrow labeled "Hydrogen bond" points to the dashed line representing the interaction between the water and ammonia molecules.

[Generated by AI]

Hydrogen bonding can also frequently be found in biological molecules. In fact, it is the force that holds the two strands of your DNA together (**Interactive 2**). We have now identified that hydrogen bonds are the strongest intermolecular force. They are so strong that to overcome this force of attraction and separate the strands you would have to heat the DNA to 95 °C or use another biological chemical known as an enzyme.

Rights of use

Interactive 2. Hydrogen bonding is found in biological molecules such as DNA.

More information for interactive 2

An interactive illustration displays the double-helix structure of DNA. It shows two long strands, which spiral around each other in a twisted-ladder shape. The strands consist of alternating sugar and phosphate groups. These strands are connected by horizontal steps that represent base pairs made of nitrogenous bases. The background features blurred DNA structures.
A plus icon, representing an interactive hotspot, is present on the structure. Selecting this hotspot reveals a schematic representation of the DNA, featuring the nitrogenous bases.
Hydrogen bonds connect the nitrogenous bases on one strand to their complementary bases on the other.
This interactive tool helps users understand how hydrogen bonds connect nitrogenous bases in DNA, contributing to its overall stability and function.

Study skills

Be aware that there can be more than one type of intermolecular force present at the same time. London dispersion forces are always present and some of the others present by themselves or all of them may be present depending on the molecule.

It is important to read all questions carefully to identify if they want you to determine all forces present or just the strongest one.

Worked example 1

The hydrogen halides do not show perfect periodicity. A bar chart of boiling points shows that the boiling point of hydrogen fluoride, HF, is much higher than periodic trends would indicate.

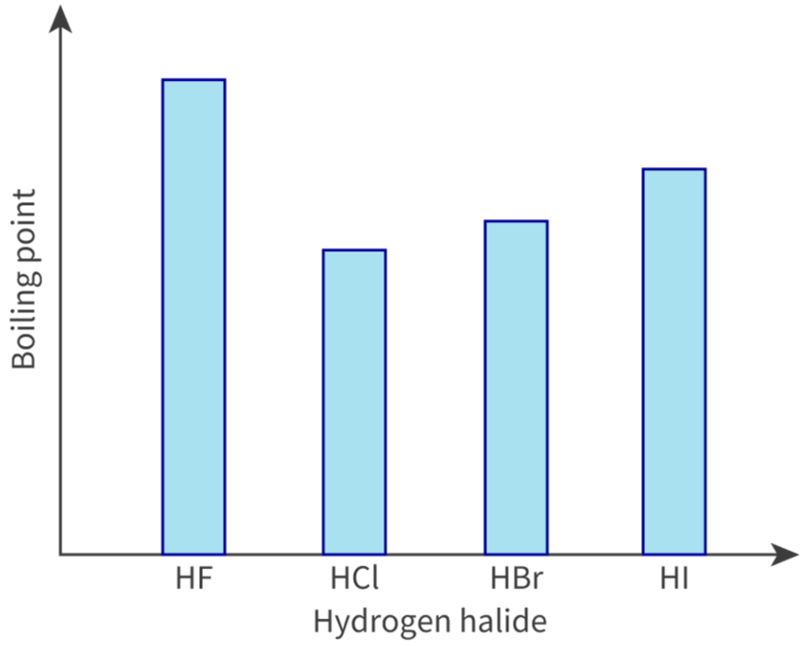

Figure 4. Boiling points of hydrogen halides.

More information for figure 4

The bar chart displays the boiling points of various hydrogen halides. The X-axis represents different hydrogen halides: HF, HCl, HBr, and HI. The Y-axis indicates the boiling point, though specific temperature units and values are not visible in the image. The chart illustrates that HF has a significantly higher boiling point compared to HCl, HBr, and HI. The bars for HCl, HBr, and HI show a trend of increasing boiling points, but none are as high as HF.

[Generated by AI]

1. Explain why the boiling point of HF is much higher than the boiling points of the other hydrogen halides.
2. Explain the trend in the boiling points of HCl, HBr and HI.

In any question to do with melting or boiling points, the first thing you need to do is work out the intermolecular forces acting between the molecules. You may need to draw them out to determine this.

1. HF has hydrogen bonds between molecules which is the strongest intermolecular force. More energy (hence higher boiling point) is required to overcome this strong force.
2. HCl, HBr and HI do not experience hydrogen bonds. All of these molecules will experience dipole–dipole forces and London dispersion forces and as you progress down the period from HCl to HI the molecular size (i.e. number of electrons) on the halogen will increase. This causes the strength of the London dispersion force to increase.

In the following activity, you will apply your knowledge of the different intermolecular forces and what causes them in two separate experimental scenarios. Interpreting and processing data, and using it to formulate conclusions are important IB inquiry skills.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
  - Thinking skills — Providing a reasoned argument to support conclusions
  - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 25 minutes
- **Activity type:** Pair activity

In your pairs, you will be given data from different experimental scenarios to analyse. You must research and explain the results together.

Then, join with another pair who researched the opposite experimental scenario. Explain your respective scenarios to each other.

Experiment scenario 1

Table 1 lists data regarding boiling points of each of the molecules that was collected in a university laboratory. Identify all of the intermolecular forces each molecule experiences. One has been done for you as an example.

Table 1. Quantitative data collected from experimental

Chemical species	Strongest intermolecular force
H_2	
Ne	
O_2	
Cl_2	
HCl	
HBr	
H_2S	London dispersion and dipole—dipole
NH_3	
HF	
H_2O	

Table 1. Quantitative data collected from experim

Chemical species	Strongest intermolecular force
H_2	London dispersion
Ne	London dispersion
O_2	London dispersion
Cl_2	London dispersion

Chemical species	Strongest intermolecular force
HCl	London dispersion and dipole—dipole
HBr	London dispersion and dipole—dipole
H_2S	London dispersion and dipole—dipole
NH_3	London dispersion and dipole—dipole and hydrogen bond
HF	London dispersion and dipole—dipole and hydrogen bond
H_2O	London dispersion and dipole—dipole and hydrogen bond

There is no electronegativity difference between molecules 1—4 so the attraction would be that created by temporary dipoles, i.e. London dispersion forces.

In molecules 5 and 7 there are polar bonds present due to the significant difference in electronegativity values. This results in dipole-dipole interactions.

Molecules 8—10 all feature hydrogens joined to a highly electronegative element and the electronegative element also has a non-bonding pair of electrons. These molecules have all met the requirements for hydrogen bonds to form.

Experiment scenario 2

The graph in **Figure 5** shows the boiling points of the group 15 hydrides. Discuss the variation in the boiling points.

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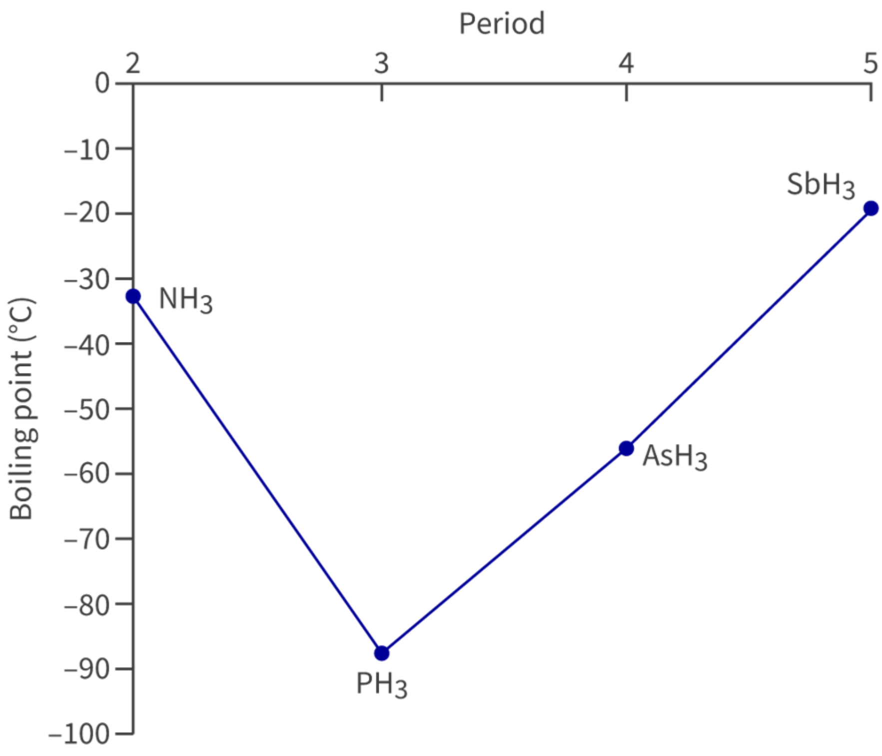

Figure 5. Boiling points of the group 15 hydrides.

More information for figure 5

The trend in boiling point is related to the strength of the intermolecular forces. PH_3, AsH_3 and SbH_3 all have dipole—dipole forces and London dispersion forces. As the mass of the molecule increases, the strength of the London dispersion forces increases, therefore increasing the boiling point. NH_3 is the only compound here that is capable of hydrogen bonding, the strongest intermolecular force, therefore it has a higher boiling point than PH_3.